

Forces			Moment of Inertia	POI	Internal Moment	
x	Braking	822,000.00	0.044400 (Area)	x	0.800	
y	Lateral	205,000.00	0.001224 I _y	y	-0.146 M _y	153,750.00
z	Vertical	1,027,000.00	0.000815 I _z	z	-0.037 M _z	770,250.00

BENDING STRESS

x	18,513,513.51
y	-6,980,061.35
z	-91,876,225.49

TOTAL -80,342,773.33

Total I -80.34

At Corner VM = O_{total}

SHEAR STRESS

	V	I	t	Q
y	205,000.00	0.001224	0.060	0.002160
z	1,027,000.00	0.000815	0.060	0.002205

HRB Dimensions

B	0.350
H	0.450
b	0.290
h	0.390

	sigma
y	9,243,865.03
z	30,205,882.35

	MPA
	9.24
	30.21

Von Mises Stress

25,493,574.86	6.50E+14	8.54E+13
84,896,164.14	7.21E+15	9.12E+14
-110,389,739.00	1.22E+16	
	2.00E+16	9.98E+14
	1.00E+16	2.99E+15
1.30E+16		
114,083,832.64		
114.08	VON MISES STRESS	